Chapter 16

An Application: Pricing Corporate Bonds

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16.1 Corporate Bonds

We now apply our methods for valuing risky payoffs to valuing corporate debt. When a corporation issues a bond, it issues a security which promises fixed future payments. We have already seen how to price such bonds in the risk free case.

$$B = \sum_t P_t X_t,$$

where P_t is the current price of getting one dollar for sure at time t, and X_t is the bond payment in period t. But this framework can not in general be applied to *corporate* bonds, because these bonds are *risky*. The CAPM type of framework is also problematic for risky bonds, due to the *contingent* nature of payoffs. The right valuation framework is therefore the state-price approach.

16.2 The Risky Part of Corporate Bonds

Corporate bonds are risky because of the default probability. To illustrate how this affects values consider the following example.

Example

The *Are You Lucky Gold Mine Company* has only two mines left. One mine has well known reserves, they are estimated to yielding a net cashflow of 1 million next period for sure, and nothing afterwards. The other mine is not yet fully explored. The cost of exploring enough to estimate the reserves is 100,000. There is a large probability the mine is empty, in which case it is worthless. There is a small probability of large gold reserves in the mine, supporting an annual production with net cashflows of 1 million per year for 5 years. The Are You Lucky Gold Mine Company has issued debt with a face value D of 1,500,000. The debt is due next period.

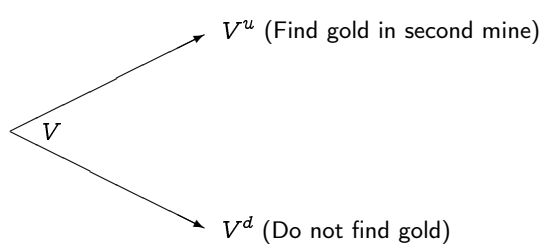
It is only in the case that the new mine is nonempty that the gold company has sufficient cash flow to pay back the debt of 1.5 million. Otherwise the firm only has assets worth 900,000 to cover 1.5 million of debt. The latter case is default, the firm goes bankrupt and its assets are taken over by the creditors, in this case the bondholders.

It is the nontrivial probability that a company goes bankrupt that makes corporate bonds risky. Since these bonds are risky, they should be valued using the by now familiar state price approach.

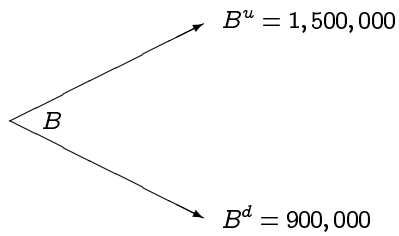
Example

What is the value to bondholders of the debt of the "Are You Lucky Gold Mine" company?

The relevant "states" are here defined by the value of the company, termed V:



What are the payments to bondholders (B)? In the "up" state the assets of the firm is worth more than the face value of debt, bondholders receive the face value. In the second case the firm is bankrupt, with total assets of 900,000 (The one million from the old mine less the cost of 100,000 of exploring the new mine).



To price we need the state price probability p^u. This could for example be estimated from the stock price of the company. Supposing $p^u = 0.48$ and $r = 0.05$, we can value the bonds of the company as

$$\begin{aligned} B &= \frac{1}{1+r}\left(p^u B^u + (1-p^u)B^d\right) \\ &= \frac{1}{1+0.05}(0.48 \cdot 1.5 + (1-0.48)0.9) = 1,131,000 \end{aligned}$$

16.3 Corporate Bonds are Riskfree Bonds with a Put Option Attached

To build further intuition about bond pricing, we can reinterpret the position for a bondholder as holding a risk free bond, but having sold a put option with exercise price equal to the face value D of the debt.

Let V' be next period's value of the firm. E is today's value of equity. A company will *default* if it does not have enough assets to pay the debtholders. Hence, the debtholders receive the following payoff pattern:

$$\min(V', D),$$

which can be re-written more lucidly as:

$$D - \max(D - V', 0).$$

This means that corporate debt is really a packet of

1. riskfree debt with face value D.
2. a *put* option that the debtholders *wrote* to the shareholders.

By the axiom of *value additivity*, we can therefore value corporate debt as the sum of the value of riskfree debt minus the value of a put option.

$$B = \frac{1}{1+r}D - \frac{1}{1+r}E^*[\max(D - V', 0)]. \tag{16.1}$$

Figure 16.1 Bond as combination of risk free bond and a short put.

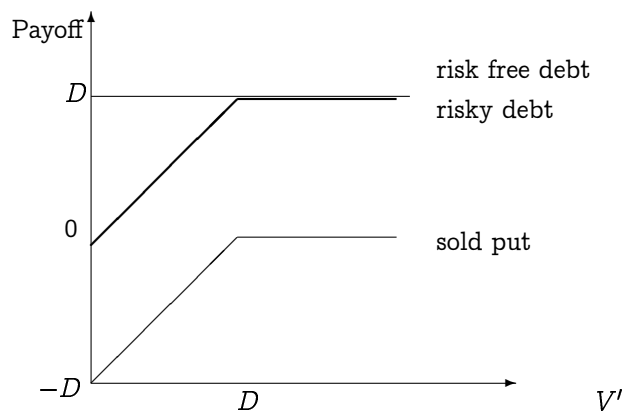


Figure 16.1 illustrates the combination.

Since equity is the residual claimant, by value additivity the value of the firm V has to equal the sum of equity and debt, $V = E + B$. That means we can *also* calculate the value of corporate debt from the value of equity E:

$$B = V - E.$$

We already discussed that E is the value of a *call option* on the assets of the firm, with payoff pattern

$$\max(V' - D, 0).$$

Corporate debt can therefore *also* be considered a combination of the assets of the firm minus a call option written to the shareholders. The valuation formula would be:

$$B = V - \frac{1}{1+r} E^* \left[\max(V' - D, 0)\right] \tag{16.2}$$

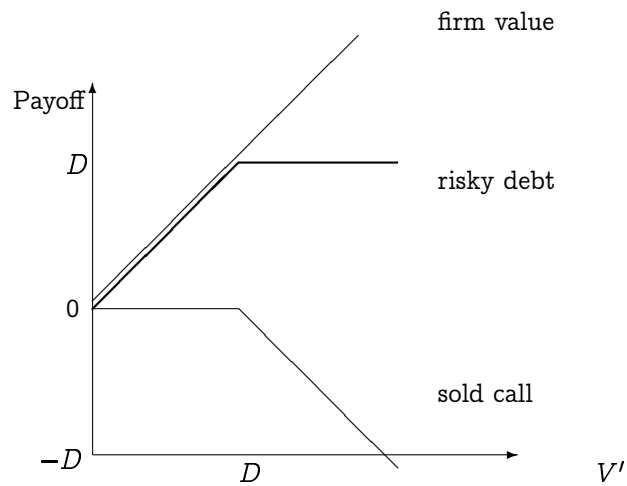
The combination is illustrated in figure 16.2.

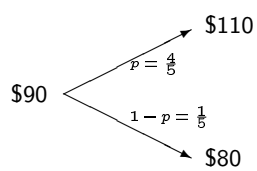
All of the approaches to pricing debt above are consistent, and either approach will give the same numerical result.

Example

A company is set up to implement a project that requires an \$80 investment outlay. The present value of the project is \$90, giving a NPV of $90 - 80 = 10$. The future value V_1 of the project is one of two: $V^u = 110$ (with probability 0.8) or $V^d = 80$ (with probability 0.2).

Figure 16.2 Bond as combination of value of the firm and a short call.





Note that these are the *true* probabilities, not the state price probabilities. We can use them to calculate the *expected return* of the project to be

$$E[r] = \frac{0.8 \cdot 110 + 0.2 \cdot 80}{80} = \frac{104}{80} = 30\%.$$

To finance the project, the management of the company decides to issue debt with a face value of \$85. The riskfree rate is 5%. We will first find how much the debt can be issued for.

To value the debt we will use relationship (16.1) above. To do so we need state price probabilities. These we can find from the value of the project, since the entire firm is also valued using the same state price probabilities:

$$V = \frac{1}{1+r} E^*[V_1] = \frac{1}{1+r} \left(p^u V^u + (1-p^u) V^d \right)$$

Plugging in numbers

$$90 = \frac{1}{1.05} \left(p^u 110 + (1-p^u) 80 \right)$$

and solving for p^u we find $p^u = 0.48$. Using these state price probabilities we can find the current value of debt as

$$\begin{aligned} B &= \frac{1}{1+r}D - \frac{1}{1+r}E^*[\max(D - V_1, 0)] \\ &= \frac{1}{1.05}85 - \frac{1}{1.05}(0.48 \cdot 0 + (1 - 0.48)(85 - 80)) \\ &= 78.5. \end{aligned}$$

The corporate debt with a face value of 85 is issued for 78.5 one period before. We can use this to find the return from buying the bond, or the *yield* on the corporate bond, as

$$\frac{D}{B} - 1 = \frac{85}{78.5} - 1 = 8.3\%.$$

The return from buying the risky debt is 3.3% higher than the 5% risk free interest rate. This reflects a *risk premium* necessary to induce anybody to be willing to buy the debt. Let us also value the debt using the alternative approach shown in (16.2) above:

$$\begin{aligned} B &= V - \frac{1}{1+r}E^*[\max(V_1 - D, 0)] \\ &= 90 - \frac{1}{1+r}(p^u \max(110 - 85, 0) + (1 - p^u)\max(80 - 85, 0)) \\ &= 78.5. \end{aligned}$$

We can use the relation between debt and equity to find the value of equity

$$B = V - E$$

giving an equity value of

$$E = V - B = 90 - 78.5 = 11.5.$$

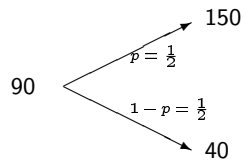
To get the company going, the shareholders will have to pay the remainder of the investment outlay, $80 - 78.5 = 1.5$. For 1.5, they get shares that are worth 11.5. The difference of 10 is of course the NPV of the project.

16.4 The Incentives for Shareholders to Change the Nature of the Assets Once the Bonds are in Place

We will now show that corporate finance is not so simple as you thought, by illustrating that the *incentives* of different groups, namely bondholders and shareholders of a given firm, may not be aligned.

Example

Suppose the management in the previous example went ahead with the 85 debt issue, yielding 78.5. In addition the firm issued shares for 1.5. Just before starting the \$80 project, management notices that a nice alternative project presents itself. The alternative project also costs 80, has the same $NPV = 10$, but the following payoff pattern:



Using the true probabilities we can calculate its expected return as

$$E[r] = \frac{0.5 \cdot 150 + 0.5 \cdot 40}{80} = 18.75\%$$

Note that the new project has a lower expected return than the alternative one. If you were to calculate the volatility, or standard deviation, of the return on the two projects, you would find that the alternative project has a higher standard deviation, it is *more variable*.

Should management switch projects? If the management wants to pick the project that maximizes the value of the firm, it would be indifferent, since both projects have the same value. If the management were (erroneously) trying to maximize expected return on value, it should stick to the old project. But managers are responsible to shareholders, so they should pick whatever project gives the highest share value. What is the share value under the new project?

We first use the NPV of the project to determine the state price probabilities under the new project, denoted $\tilde{p}^u$

$$V = 90 = \frac{1}{1.05}\left(\tilde{p}^u 150 + (1 - \tilde{p}^u)40\right),$$

giving a state price probability $\tilde{p}^u = 0.50$. Use $\tilde{p}^u$ to determine the value of equity:

$$E \quad = \quad \frac{1}{1+r}\left(\tilde{p}^u(150 - 85) + (1 - \tilde{p}^u)\,0\right) = \frac{1}{1.05}0.50 \cdot 65 = 31$$

The share value under the new project is 31. Comparing this to the share value under the original project of 11.5, the management should clearly switch to the new, more risky project.

Switching projects really amounts to "expropriating" the bondholders. If the bondholders had *known* the company would use the raised capital to invest in the alternative, more risky project, they would only have been willing to put up

$$B = V - E = 90 - 31 = 59$$

to finance the project. That is, the bondholders would demand an increase in the *yield* of the bond to

$$\frac{D}{B} - 1 = \frac{85}{59} - 1 = 44\%$$

to compensate for the increased risk of default. But since the bonds have already been issued at 78.5, management has the cash on hand... So, yes, the management should switch projects, and the debtholders bear the cost of the increased riskiness.

In fact, this is a fairly general result. After the debt is in place, management always has an incentive to increase the riskiness of the firm, thereby increasing the value of equity, while reducing the value of debt.

Debt covenants are meant to keep the management from doing so. Or, alternatively, the management could attach *warrants* to the bonds (see Chapter 13), or issue *convertible bonds*. Let us discuss the latter.

16.5 A Solution: Convertible Bonds

A convertible bond gives the holder the right to exchange the bond for a number of shares of equity. As such they combine features of bonds with features of warrants.

Example

A corporation has issued convertible bonds with a face value of 1000. At the discretion of the bondholder each bond can be exchanged for $33\frac{1}{3}$ shares of newly issued common stock.

The *conversion ratio* of a convertible bond is the specification of how many shares are issued *per bond*. The *conversion price* is the price of the underlying stock which makes the face value of the convertible bond equal to the value of the converted bond.

Example

In the previous example the conversion ratio is $1 : 33\frac{1}{3}$ and the conversion price is 30.

Convertible bonds create problems that are similar to those for warrants. "Dilution" is a confusing concept, and optimal conversion policies are difficult to determine. We will avoid problems created by conversion and refer a study of them to more advanced texts. Let us look at an example where there is *one* share and *one* convertible bond, and the conversion ratio is 1:1. The notation is the same as earlier in this chapter.

Then, the terminal payoff on the convertible is as follows.

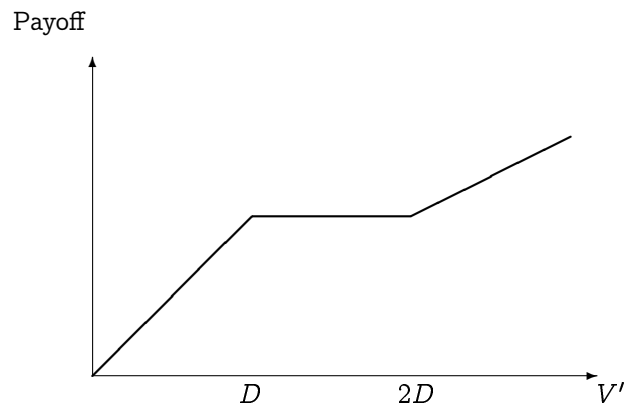
1. If $D < V' < 2D$, then the payoff is D.
2. If $V' \leq D$, then the company defaults and the payoff is V'.
3. If $2D \leq V'$ then it pays to convert the bond and receive the value of one share, which is $V'/2$.

Graphically, the payoff pattern is shown as in figure 16.3.

A convertible bond can be valued as the combination of

1. a risk free bond.

Figure 16.3 Payoff for Convertible Bond

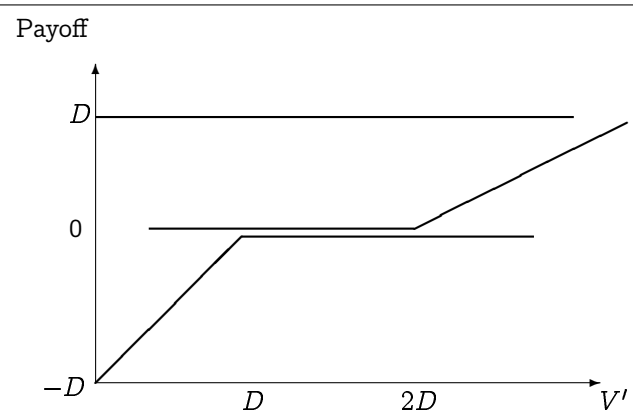


2. (writing) a put option.

3. a call option on the shares of the firm.

Figure 16.4 illustrates the combinations

Figure 16.4 Convertible bond as combination of straight bond, put and call.



With value additivity, this gives the following valuation formula:

$$B = \frac{1}{1+r}D - \frac{1}{1+r}E^*\left[\max\left(D - V', 0\right)\right] + \frac{1}{1+r}E^*\left[\max\left(\frac{V'}{2} - D, 0\right)\right].$$

16.6 Callable Convertible Bonds

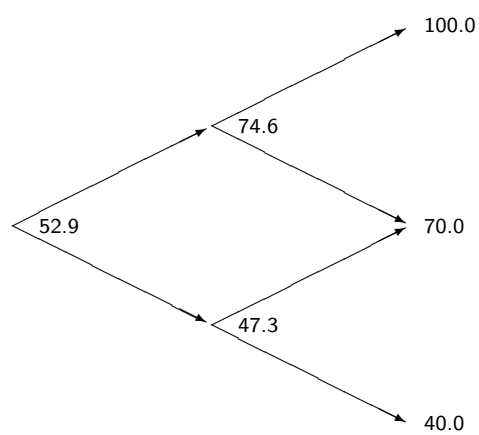
Many firms attach a *call provision* to convertible bonds. That is, the firm reserves the right to call back the bond issue at a pre-set *call price*. When called, bondholders do have the right to convert the bonds to equity instead of delivering the bonds against the call price. Let $\overline{C}$ denote the call price. An interesting question is: *when should the management call the convertible bonds?* The answer is that whatever policy the management uses, it should be in the interest of the shareholders. Since

$$E = V - B$$

and (unlike warrants) the value of the firm is unaffected by conversion, E is maximized by minimizing the value of the convertible bonds. That will happen if the management chooses the conversion policy that minimizes the value of the conversion option. How? By calling as soon as the conversion option is *in the money*, that is, the conversion value rises above the call price. The result of the call is obviously that the bondholders will convert, because the conversion value is at least as large as the call price. The mechanics and the effect on valuation are best understood with an example.

Example

We look at a situation with three dates, $t = 0$, $t = 1$ and $t = 2$. The riskfree rate per period is 10%. The evolution of the firm's value at t, V_t, can be described by a binomial tree:

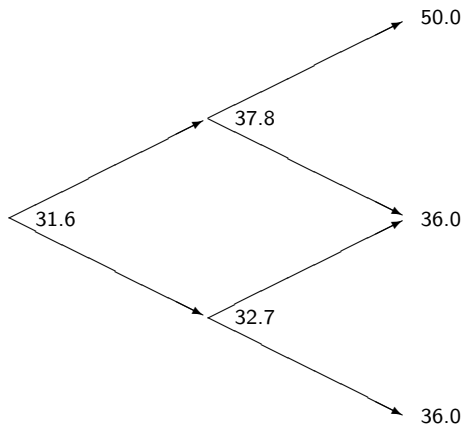


These numbers were chosen such that the (per period) state-price probability is always the same

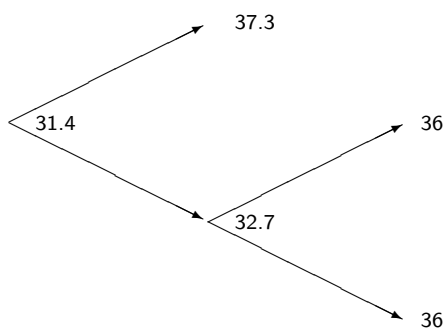
$$p^u = 0.40.$$

There is one share of equity and one convertible bond. The bond has a face value of $D = 36$, to be paid at $t = 2$. There is no intermediate coupon payment. The bond is callable *only at* $t = 1$ for $\overline{C} = 37$. The conversion ratio is 1:1.

Ignoring the call provision, the bond can be valued by recursively applying our valuation formula (10.1) along the branches of the binomial tree. The result is:



Now consider the call feature. It is optimal to call when the continuation value (of the bond) is more than the call price $\overline{C} = \$37$. This is the case in the "up"-state at $t = 1$. Notice, however, what happens. In that state, the conversion value is \$37.3 (total firm value \$74.6, divided by the two shares that will exist after conversion). So, rather than tendering their holdings to the firm and get only \$37, the bondholders will convert and receive a share worth \$37.3. That is, *the call effectively forces conversion.* In the "down"-state, nothing changes, because the continuation value of the bond is below the call price. The resulting effect on today's value of the convertible bond is:



Note that the continuation of the tree in the "up"-state at $t = 1$ is cut off, the tree is "trimmed" because of the firm's exercise policy.

References

See Smith (1976), Smith (1979) and Cox and Rubinstein (1985) on convertible bonds. See Myers (1977) on investment distortions. See also the recent summary by Myers (2001).

Problems

16.1 *Conversion* [1]
Why does conversion of convertible bonds not affect the value of the firm?

16.2 *Bond Covenants* [3]
In one or two sentences, answer the following.

1. Who benefits from the covenants in bond contracts when the firm is in financial trouble? Why?
2. Who benefits from the covenants in bond contracts when the firm is issuing debt? Why?

16.3 [3]
The Q corporation will next period realize a project that will have value either 100 or 20. This project is the only assets that Q corporation have. Q has issued a bond with face value of 50, due next period. The risk free interest rate is 10% and the current value of equity in Q is 40. Determine the current value of the bond.

16.4 *Projects* [7]
A start-up company considers two investment projects that require a \$90 investment. Both are zero NPV projects (hence, the value of the project is \$90). Only one project can be implemented. Part of the necessary funds are to be acquired through a zero-coupon debt issue, with face value of \$85. The remainder is collected through an equity issue. The specifics of the two projects are

- Project A:

$$\text{Final value } = \begin{cases} 110 & \text{with probability } 0.8 \\ 80 & \text{with probability } 0.2 \end{cases}$$

- Project B:

$$\text{Final value } = \begin{cases} 150 & \text{with probability } 0.5 \\ 40 & \text{with probability } 0.5 \end{cases}$$

1. Which project has the highest expected return?
2. Assume management randomly picks a project, and decides to choose Project A. (Management should not have any particular preference, because both projects have a zero NPV.) How much money will the company raise from the debt issue? How much equity will be raised? Assume that the risk free interest rate equals 5%.

3. Assume now that the debt and equity issues to finance project A has been completed. Now the management turns around and does not implement project A, but, instead, goes ahead with project B. What is the loss in value to the debtholders? Would the equityholders applaud this move?

16.5 *Convertible* [6]

Suppose the firms end of period value will be:

$$\text{Value} = \begin{cases} 1500 & \text{with probability } 0.6 \\ 800 & \text{with probability } 0.4 \end{cases}$$

Today's firm value is 1000. The risk free rate is 5%. The firm has 100 shares of equity and 100 convertible bonds with a face value of 10 each. The bond pays no coupon. One bond can be converted into one share.

1. Compute the value of the convertible bond and of the equity.

16.6 *Yazee* [4]

Yazee is valued at $V = 100$. Tomorrow's value V', will be either 150 or 50, with equal chance, Yazee has issued a corporate bond with face value 100 and no coupon, to be paid tomorrow. The bonds presently give a 15% yield. Is the bond mispriced? The risk free rate is 10%.

16.7 *AoB* [4]

AOB, Inc., has issued 2 shares of common stock and 1 convertible bond. AOB is valued at $V = 100$. Tomorrows value, V', will be either 150 or 50, with equal chance. The bond has a face value of \$30 and carries a coupon of 10%. The conversion ratio is 1:1. The bond is due tomorrow. The risk free rate is 10%. Value the bond.